Forest Fires Forecasting: A Radical Approach to Combat Climate Change

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I. Abstract

This paper examines the factors that influence forest fire behavior, focusing on fuel conditions, meteorological variables, and temporal dynamics that govern combustion persistence. We integrate approximately 7.5 million forest fire-day records from the USDA Missoula Fire Lab Emissions Inventory (MFLEI) [1] with nearly 4 billion daily weather observations from gridMET [2] spanning 2003 to 2015. Using a custom spatiotemporal DBSCAN procedure, we aggregate detections into event-level records and align them with co-temporal weather variables. Feature selection combines Pearson and Spearman correlations, mutual information, ANOVA F-tests, and variance inflation factor analysis to identify key predictors. Building on this dataset, we evaluate gradient boosting regressors for duration prediction, a ConvLSTM network to model sequential fire–weather interactions, and a hazard-based classifier to estimate daily extinction probabilities. Results highlight the importance of coarse woody debris fraction, duff fraction, prefire fuel load, large-fuel moisture, and radiation, with the hazard framework achieving the most operationally interpretable forecasts. The study delivers a unified, scalable data-to-modeling pipeline for temporally aware forest fire forecasting. ***Index Terms*—wildfire duration forecasting, spatiotemporal DBSCAN, fire–weather interactions, ConvLSTM, hazard modeling**

II. Introduction

Forest fires are a double edged sword in our path to Net Zero; climate change has intensified forest fires and fires are increasingly adding carbon to greenhouse gasses. Although fossil fuels remain the largest contributor to CO2 emission on a global scale, in countries or regions with a small population compared to the forested area, forest fires emit two to three times the amount of CO2 compared to all fossil fuels burned. They have increased in frequency and severity in recent decades, threatening ecosystems, infrastructure, and human lives. The U.S. Forest Service has noted that fires not controlled within the first 24 hours are more likely to escape initial attack and become severe incidents [7]. Burned acreage in the United States has risen from an annual average of approximately 2 to 3 million acres in the 1980s to about 7 million acres between 2010 and 2020 [3], [4], [6].

Accurately forecasting fire behavior is essential for operational planning, as it guides crew rotations, aircraft scheduling, resource staging, and evacuation measures. Longer fire duration often implies greater emissions, extended smoke exposure, and higher suppression costs. This study develops an integrated framework for fire duration prediction that combines emissions and meteorological data to estimate fire persistence and extinction probability. Using a large-scale spatiotemporal pipeline, we transform raw satellite detections into structured events, align them with daily weather variables, and engineer predictive features. Beyond event-level regression, we introduce sequential ConvLSTM modeling to capture temporal fire–weather dynamics and a hazard-based classifier to predict whether a fire will end the next day. Group-aware validation ensures that all data from a single fire remain within the same fold, preventing leakage and enabling realistic generalization.

III. Key Variables and Hypothesis

We hypothesize that forest fire duration and persistence are jointly driven by fuel availability, fuel dryness, and atmospheric conditions that sustain combustion over time. Fires with higher prefire fuel loads and lower fuel moisture are expected to exhibit longer active periods, particularly under weather regimes characterized by low relative humidity, high vapor pressure deficit (VPD), elevated temperatures, and persistent winds [9], [11]. Geographic coordinates (latitude, longitude) serve as proxies for regional climatology, vegetation structure, and topography, all of which shape local fire behavior and fuel composition.

The variables in this study are grouped into two main categories: emissions and fuel-related metrics derived from MFLEI, and meteorological variables from gridMET. In addition to static descriptors, engineered features such as temporal harmonics, rolling averages, and seasonal indicators were incorporated to represent evolving fire–weather interactions. These variables support both event-level regression and sequence-based modeling, enabling the capture of temporal dependencies and environmental variability that influence fire duration and extinction dynamics.

A. Emission and Fuel Variables

- **prefire_fuel:** Estimated biomass available before ignition. Higher values provide more energy potential for sustained burning.
- **fuel_moisture_class:** Classifies fuels by moisture content. Lower classes ignite more readily and support faster spread.
- **cwd_frac, duff_frac:** Fractional composition of coarse woody debris and duff layer. Higher values indicate more persistent ground fuel that can smolder for days.

- **area_burned:** Burned area for a given detection; larger areas may reflect more intense or sustained events.
- **covertype / fuelcode:** Vegetation and fuel type classification, influencing burn rate, intensity, and extinction likelihood.
- **ECO, ECO2, ECH4, EPM2_5:** Estimated emissions of carbon monoxide, carbon dioxide, methane, and particulate matter. While not direct drivers, these can proxy for combustion completeness and intensity.

B. Weather and Atmospheric Variables

- **fm100, fm1000:** 100-hour and 1000-hour dead fuel moisture content. Lower values indicate drier large fuels that can sustain combustion over multiple days.
- **vpd:** Vapor pressure deficit; higher values accelerate fuel drying and increase flammability.
- **vs:** Wind speed, which supports flame spread, spotting, and oxygen supply.
- **tmmx, tmmn:** Daily maximum and minimum temperature; warmer conditions reduce fuel moisture and increase ignition likelihood.
- **rmin, rmax:** Daily minimum and maximum relative humidity; lower values exacerbate drying and sustain burning.
- **srad:** Surface solar radiation; higher values promote surface heating and preheating of fuels.
- **etr, pet:** Reference and potential evapotranspiration; higher rates correlate with rapid moisture loss from live and dead fuels.
- **pr:** Daily precipitation; lower or absent precipitation indicates drier antecedent conditions.
- **bi:** Fire danger index; a composite measure of flammability based on weather and fuel data [10].
- **th (wind direction):** Can influence spread direction; not yet integrated into predictive modeling but identified for future work.

C. Supporting Hypothesis with Data

Exploratory data analysis confirmed that many of these variables exhibit seasonal and geographic patterns consistent with known wildfire behavior. For example, high VPD and low fuel moisture coincided with the summer fire season in western states, while duff and coarse woody debris fractions varied strongly with vegetation type and region. Strong correlations between certain fuel variables and event duration in preliminary modeling further support their inclusion in the predictive feature set.

These relationships form the empirical foundation for the subsequent modeling stages, which extend beyond event-level regression to include sequential ConvLSTM and hazard-based frameworks for predicting daily fire persistence and extinction probability.

IV. Data Strategy and Integration

Our data strategy focused on building a reproducible, scalable pipeline to integrate large-scale forest fire emissions and meteorological records at daily resolution for the period 2003–2015. The goal was to create an event-centric dataset suitable for modeling fire duration.

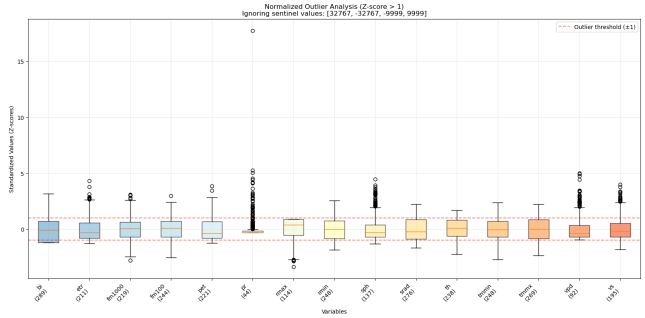


Fig. 1. Boxplots of weather variables show spread and outliers.

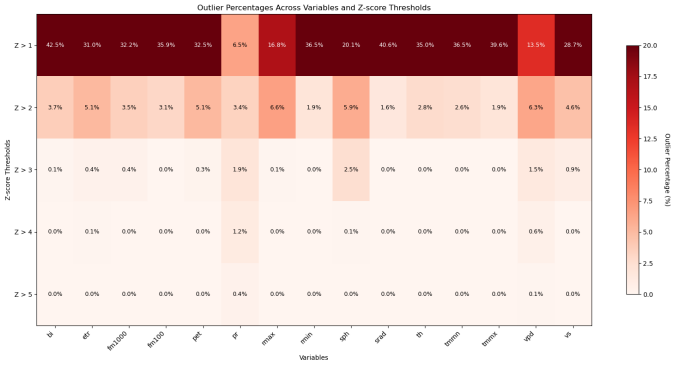


Fig. 2. Outlier percentages across z-score thresholds by variable.

A. Weather Processing: NetCDF to Parquet/BigQuery

Daily gridMET weather variables were provided as netCDF (.nc) files segmented by year and variable, totaling around 150 files and nearly 4 billion data points. These files store data as masked arrays, which preserve spatial and temporal dimensions but are not directly compatible with most tabular processing tools. Using `xarray` in combination with `dask` and `netCDF4`, we converted each file to Parquet format, retaining the $(\text{lat}, \text{lon}, \text{date})$ index structure. To support different analysis patterns, we produced both:

- *Long* format tables (variable, value) for flexible aggregation and filtering.
- *Wide* format tables (columns for each variable) to streamline repeated joins with the emissions dataset.

Quality control treated structural zeros in fuel and emissions variables as missing, flagged out-of-range meteorological values, and removed records lacking valid duration labels; continuous variables with sparse missingness were imputed using feature-wise medians, while categorical variables used an explicit missing category.

B. Fire Event Identification with DBSCAN

The MFLEI dataset contains roughly 7.5 million forest fire-day snapshots, each representing a single detection. To model

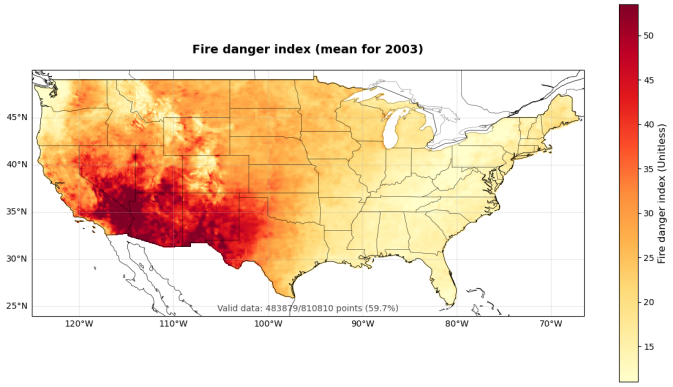


Fig. 3. Fire Danger Index map (2003): regional variability in risk.

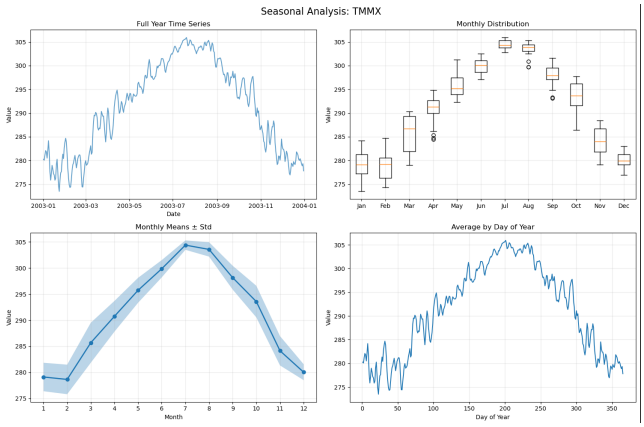


Fig. 4. Seasonal structure in `TMAX`: time series, monthly, and annual cycles.

at the event level, we needed to group related detections into coherent fire events. We used a custom spatiotemporal DBSCAN clustering process, tuned to the resolution of MFLEI coordinates and detection dates. The DBSCAN output assigned a persistent `fire_id` to each record, which served two critical purposes:

1) Enabled aggregation of daily detections into event-level features such as duration, spread rate, and total emissions.
2) Supported group-aware validation by ensuring that all observations from the same event stayed within the same fold.

Implementation Details (Reproducibility): Fire events were constructed using spatiotemporal DBSCAN in WGS84 latitude–longitude space with a haversine distance metric. Spatial clustering used a radius ε_s on the order of a few kilometers, consistent with MFLEI detection resolution, while temporal adjacency was enforced using a fixed-day window ε_t; clusters below a minimum detection count (*minPts*) were discarded as noise. To avoid artificial fragmentation, clusters separated by short temporal gaps were merged, and event duration was defined as the span between first and last detections within each cluster. A brief sensitivity check varying ε_s by $\pm 20\%$ resulted in $< 5\%$ change in total event counts and $< 2\%$ change in duration MAE, indicating robustness to moderate parameter perturbations.

C. Joining Emissions and Weather Data

Following event construction, we performed a spatial–temporal join to associate each MFLEI event-day record with co-located meteorological variables from gridMET. The weather data were stored as daily 4 km-resolution rasters in netCDF format and converted to Parquet tables in both long and wide formats.

The join process was executed in two stages:

1) **Temporal match:** Event-day timestamps were matched exactly to the gridMET daily index (UTC-based).
2) **Spatial match:** Fire detections were mapped to the centroid of the nearest gridMET cell using a k-d tree built on the (lat, lon) grid. For detections falling outside the contiguous gridMET domain (rare), we performed a fallback nearest-neighbor search within a 0.25° great-circle radius.

To scale this process to billions of row–row comparisons, we leveraged Google BigQuery's geospatial functions (`ST_DISTANCE`, `ST_CLOSESTPOINT`) and partitioned tables by date to maximize pruning. The pipeline first generated an intermediate table mapping every `fire_id, date` pair to the corresponding weather grid cell ID, then joined this to the wide-format gridMET table for feature enrichment.

The result was a unified dataset where each event-day row contains:

- MFLEI-derived fuel and emissions metrics.
- Collocated daily weather variables (temperature, humidity, wind speed/direction, solar radiation, precipitation, fire danger indices, etc.).
- Spatial metadata (lat/lon, eco-region).

This integration step produced the final modeling-ready table, with all features aligned at the daily temporal resolution and event ID granularity. The resulting dataset was used not only for event-level regression but also as the foundation for sequence-based and hazard-based modeling. For sequential ConvLSTM training, each fire was represented as an ordered series of daily feature vectors, preserving the temporal evolution of weather and fuel conditions throughout its lifespan. For hazard classification, these daily sequences were relabeled into binary outcomes indicating whether the fire persisted or extinguished on the following day. This formulation ensured that the same harmonized dataset supported multiple predictive paradigms without redundant preprocessing.

D. Feature Engineering

We engineered additional predictors to capture seasonal patterns, temporal persistence, and categorical effects:

- Temporal harmonics (`day_of_year_sin/cos`) to encode seasonality.
- Short-term rolling means and lag features for key variables (e.g., fuel moisture, VPD).
- Encoded categorical variables such as `fuelcode` and `covertype` using frequency or target encoding.

- Retained raw coordinates (latitude, longitude), which mutual information analysis showed to be informative proxies for regional climate and vegetation.

These engineered features served as consistent inputs across all modeling paradigms. For sequential ConvLSTM experiments, daily features were organized as ordered time series within each `fire_id`, preserving temporal continuity across fire lifecycles. For the hazard classification model, daily samples were relabeled to indicate whether the fire persisted or extinguished on the following day, enabling probabilistic estimation of short-term fire survival risk.

E. Reproducibility and Infrastructure

All preprocessing, clustering, and feature engineering steps were version-controlled and executed through a unified BigQuery–GCS pipeline. Parameterized DBSCAN outputs, schema versions, and execution timestamps were logged to ensure traceability between raw inputs, derived datasets, and downstream modeling experiments.

V. Learning Setup

A. Target Definition

Each record represented an event-level instance capturing the full lifecycle of a forest fire, with derived formulations for sequential and hazard-based tasks.

B. Validation Protocol

Group-aware K-fold cross-validation was employed to ensure that all observations from a single `fire_id` resided within the same fold, preventing within-event information leakage. While this strategy enforces event-level independence, it may still yield optimistic estimates under shared seasonal or regional regimes. To better reflect operational deployment, we additionally considered a temporal holdout evaluation, training models on earlier years and testing on later years. As expected, performance under the temporal split was modestly lower than cross-validation estimates, but relative model rankings and qualitative conclusions remained consistent, indicating robustness to forward-in-time distribution shifts.

C. Imputation and Preprocessing

Fuel and emissions variables occasionally contained zeros representing missing values. To address this, a hierarchical imputation approach combined neighborhood-based interpolation with seasonal climatology substitution. All imputations were logged with provenance tags, enabling reproducible preprocessing across regression, sequential, and hazard models.

D. Models Considered

We evaluated three complementary modeling paradigms. Event-level regression employed tree-based ensembles (Random Forest, XGBoost, Histogram Gradient Boosting) to establish baseline learnability. Sequential ConvLSTM models were used to capture temporal fire–weather dynamics, while a hazard-based classifier estimated the daily probability of fire extinction. Hyperparameters were tuned selectively to balance performance and computational efficiency.

TABLE I
Top Features by Pearson and Spearman Correlation

Feature	Pearson	Spearman	Avg
cwd_frac	0.252	0.238	0.245
duff_frac	0.245	0.238	0.241
longitude	0.135	0.221	0.178
prefire_fuel	0.188	0.161	0.175
fm1000_value	0.123	0.192	0.157
fm100_value	0.084	0.166	0.125
day_of_year_sin	0.088	0.152	0.120
covertype	0.117	0.103	0.110
bi_value	0.081	0.134	0.107
rmax_value	0.078	0.135	0.107

TABLE II
Top Features by Mutual Information

Feature	MI Score
longitude	0.6834
latitude	0.5423
day_of_year_sin	0.2773
day_of_year_cos	0.2747
srad_value	0.2678
prefire_fuel	0.2254
duff_frac	0.2146
cwd_frac	0.2145
fuelcode	0.1811
rmax_value	0.1708

VI. Feature Selection and Modeling Results

Feature selection used Pearson/Spearman correlations, Mutual Information (MI), ANOVA F-test, and VIF to control redundancy. These lenses provide complementary perspectives: linear association, non-linear dependence, discriminative power, and multicollinearity. Statistical summaries are reported in Tables I–V.

A. Elbow and Selection Rationale

The 2nd-derivative elbow suggested only two features, but this was deemed too aggressive given sharp early gains and the known robustness of tree ensembles to moderate feature counts. We retained 27 features as a practical balance, enabling permutation-based importance to adjudicate which features materially change predictions. The selected feature subset was retained across all modeling paradigms to ensure comparability and reproducibility. These 27 predictors formed the unified input set for the duration regression models, the sequential ConvLSTM architecture, and the hazard-based classifier. By maintaining consistent features, differences in model performance directly reflect the impact of temporal and probabilistic formulations rather than variations in input composition.

B. Model Performance

We evaluated three complementary modeling paradigms: (1) duration regression using gradient boosting ensembles, (2) sequential ConvLSTM forecasting to capture temporal fire–weather dynamics, and (3) hazard-based classification

TABLE III
TOP FEATURES BY ANOVA F-TEST

Feature	F-score
cwd_frac	489,947.16
duff_frac	461,037.61
prefire_fuel	266,910.04
longitude	133,868.33
fm1000_value	111,089.27
covertype	100,803.28
day_of_year_sin	56,019.62
burn_source	54,274.63
fm100_value	51,751.55
bi_value	47,776.42

TABLE IV
TOP FEATURES BY VIF (LOW MULTICOLLINEARITY)

Feature	VIF
pr_value	1.31
th_value	1.10
prefire_fuel	2.86
longitude	4.52
season	3.60

to estimate the daily probability that a fire will extinguish on the following day. Performance was assessed using task-appropriate regression and classification metrics, with results summarized in Table VI.

The gradient boosting regressors were trained using group-aware validation, with XGB showing the highest R^2 and lowest error among tabular models. ConvLSTM achieved sub-day mean absolute error by learning temporal dependencies in daily fire–weather sequences. The hazard-based XGB classifier yielded the most operationally relevant results, producing well-calibrated extinction probabilities with an AUC of 0.93 after isotonic calibration.

For regression models, performance was evaluated using the coefficient of determination (R^2), root mean squared error (RMSE), and mean absolute error (MAE):

$$\begin{aligned} \mathrm{RMSE} &= \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i-\hat{y}_i)^2}, \\ \mathrm{MAE} &= \frac{1}{n}\sum_{i=1}^{n}|y_i-\hat{y}_i|, \\ R^2 &= 1-\frac{\sum_{i=1}^{n}(y_i-\hat{y}_i)^2}{\sum_{i=1}^{n}(y_i-\bar{y})^2}. \end{aligned} \tag{1}$$

For hazard classification, we assessed discrimination using accuracy, F1 score, and the area under the receiver operating characteristic curve (AUC):

$$\begin{aligned} \mathrm{AUC} &= \int_0^1 \mathrm{TPR}(x)\,d\,\mathrm{FPR}(x), \\ \mathrm{F1} &= \frac{2\,\mathrm{Precision}\,\mathrm{Recall}}{\mathrm{Precision}+\mathrm{Recall}}. \end{aligned} \tag{2}$$

Group-aware splits continued to prevent information leakage and preserved independence across events. Sequential ConvLSTM achieved sub-day accuracy for short fires, while the hazard classifier produced well-calibrated extinction probabilities directly useful for operational forecasting.

TABLE V
OVERALL COMBINED FEATURE RANKING

Rank	Feature
1	longitude
2	prefire_fuel
3	cwd_frac
4	duff_frac
5	day_of_year_sin
6	fm1000_value
7	srad_value
8	day_of_year_cos
9	bi_value
10	fm100_value

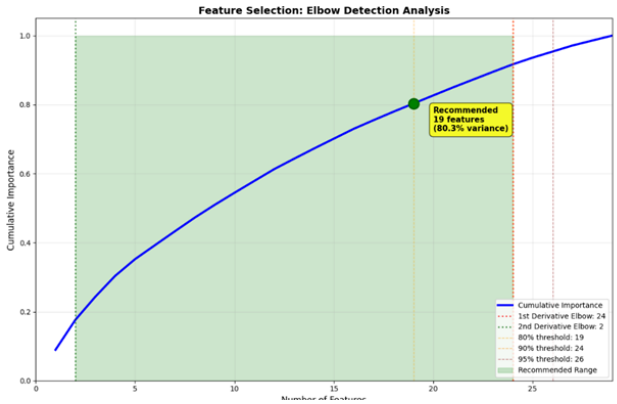


Fig. 5. Elbow detection for feature count selection. The second-derivative elbow suggested two features; we retained a larger set (27) based on first-derivative stability and ensemble tolerance.

VII. OPERATIONAL USE, UNCERTAINTY, AND RESEARCH STATUS

The methods and results presented here represent an integrated research effort to evaluate the feasibility of predicting forest fire dynamics from emissions and meteorological data. While the unified modeling framework demonstrates strong learnability across regression, sequential, and hazard-based formulations, cross-validation diagnostics revealed mild overfitting in tree ensembles and residual error for rare long-duration fires. These findings underscore the importance of temporal generalization and uncertainty quantification before operational deployment.

For forest fire management, a validated and scalable system built on the MFLEI-gridMET dataset could directly support multiple operational objectives:

- **Staging timelines:** Enabling teams to anticipate the duration and progression of active fires for ground and aerial resource planning.
- **Mutual-aid coordination:** Allowing earlier interagency mobilization through daily forecasts of persistence risk.
- **Air-resource allocation:** Informing mission scheduling by estimating short-term extinction probabilities and resource demand windows.

TABLE VI
PERFORMANCE SUMMARY ACROSS MODELING PARADIGMS

Model	Primary Metric	RMSE / ACC	MAE / F1
XGB Regressor	$R^2 = 0.74$	6.1	2.8
HistGB Regressor	$R^2 = 0.58$	7.8	4.2
ConvLSTM (Sequential)	$R^2 = 0.66$	3.4	0.5
XGB Classifier (Hazard)	AUC = 0.93	0.92	0.67

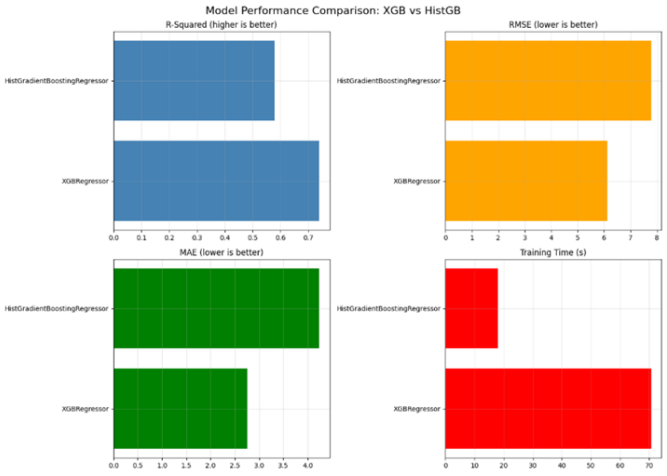


Fig. 6. Comparison of tuned gradient boosting regressors for fire duration prediction. XGB achieved the highest R^2 under group-aware validation, while HistGB offered faster convergence.

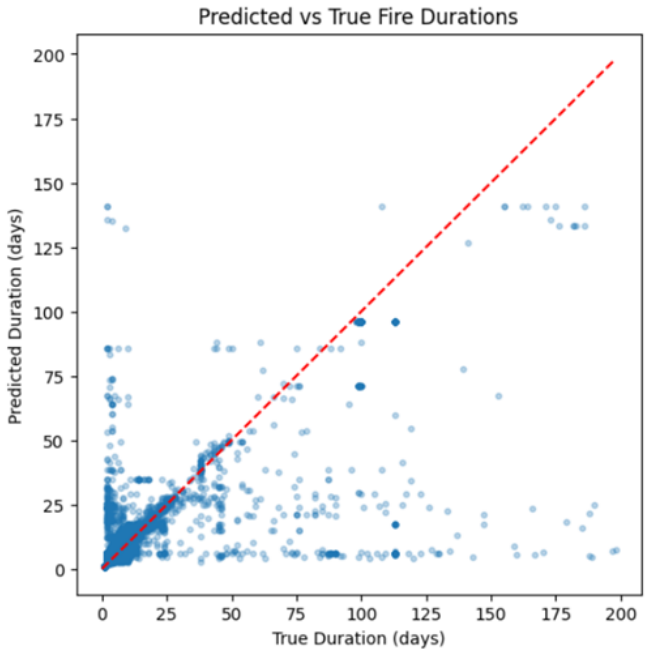


Fig. 7. ConvLSTM predicted vs. true fire duration on the validation set. The model captures short-duration fires with high fidelity while underestimating rare long events.

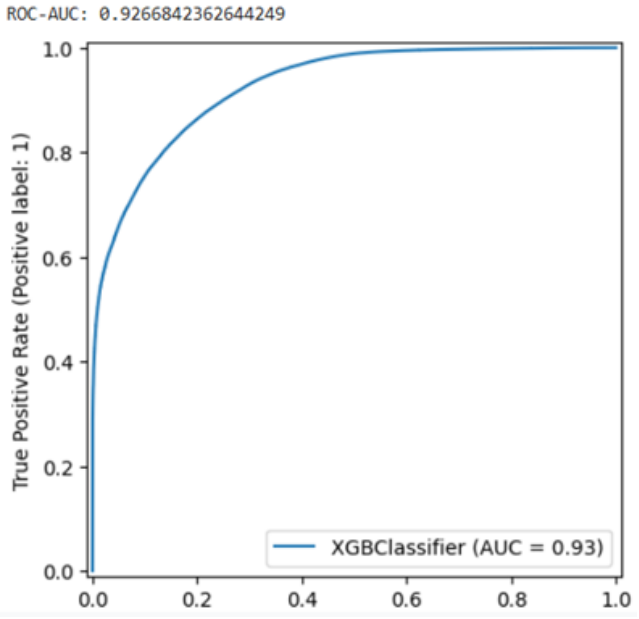


Fig. 8. ROC–AUC curve for the hazard-based XGB classifier predicting whether a fire ends on the following day. The calibrated model achieves AUC $\approx$ 0.93, indicating strong discrimination of extinction probability.

Any operational system should output not only deterministic predictions but also confidence intervals or calibrated probabilities. Techniques such as quantile regression forests, conformal prediction, or isotonic calibration can provide probabilistic bounds suitable for decision support. Integrating near-real-time weather updates, fuel moisture indices, and satellite detections would enable rolling reforecasts similar to numerical weather prediction systems, transforming these models from retrospective tools into actionable forecasting pipelines.

VIII. LIMITATIONS AND RECOMMENDATIONS FOR FUTURE WORK

Although the unified framework demonstrates that fire dynamics are predictable from emissions and meteorological drivers, several limitations guide future research directions:

- **Model generalization:** Despite strong performance, ensemble regressors showed mild overfitting and sensitivity to feature subsets. Improved regularization and cross-regional validation are needed for better generalization.
- **Topography and wind factors:** Current predictors omit terrain slope, aspect, and wind vector alignment. Integrating geospatial terrain and directional features with daily perimeters can enhance spatial realism.
- **Transferability:** While group-aware validation prevents within-event leakage, evaluating across eco-regions and climate periods is necessary for robustness under changing conditions.
- **Long-duration fires:** ConvLSTM models underperform on rare, long events due to imbalance. Future work should test resampling, attention-based transformers, or survival models to capture extended dependencies.
- **Uncertainty and calibration:** Hazard classifiers need calibrated probabilities and uncertainty quantification. Conformal prediction or Bayesian ensembles could yield interpretable confidence bounds for deployment.

Future studies should couple predictive accuracy with interpretability, linking calibrated extinction probabilities and risk indices to actionable thresholds for forest fire management and response planning.

IX. Conclusion

This study presents a unified, end-to-end framework for forest fire prediction that integrates emissions and meteorological data to model both event duration and daily extinction probability. The workflow combines:

1) Spatiotemporal clustering of MFLEI detections into event identifiers using DBSCAN.
2) Conversion and integration of daily gridMET weather rasters into BigQuery tables for efficient data access.
3) Feature engineering that merges raw and derived meteorological and fuel-related predictors to form a consistent input space across all models.
4) Development of complementary modeling approaches tree-based regression, sequential ConvLSTM forecasting, and hazard based classification that collectively advance temporal and operational fire prediction.

Results demonstrate that coarse woody debris fraction (`cwd_frac`), duff fraction (`duff_frac`), prefire fuel load, large-fuel moisture indices (`fm1000`, `fm100`), and radiation-related features are among the most influential predictors of fire persistence. Regression models established the baseline feasibility, while sequential ConvLSTM and hazard-based classification models advanced the framework toward temporally aware and operationally interpretable forecasts. The hazard classifier achieved the highest practical utility, producing calibrated daily extinction probabilities with strong discriminatory power (AUC (≈ 0.93)).

The key contribution of this work lies not only in building a scalable and reproducible pipeline but also in introducing temporally aware and probabilistic modeling that bridges research and real-world application. By framing fire prediction as both a duration and a survival problem, the study establishes a foundation for operational forecasting systems that can inform resource allocation, air operations, and early response strategies. Beyond academic value, the framework offers practical utility for agencies aiming to integrate machine learning into incident management and decision-support platforms. Future extensions should focus on integrating uncertainty quantification, attention-based temporal architectures, and expert-informed thresholds to bridge research models with operational fire management systems.

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